

Problem Sheet: Chain Rule and Substitutions 1

Solution

Jeffrey Thompson

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Exercise 1. Differentiate the following functions.

1. $\ln(5x)$
2. e^{-20x^2}
3. $\sin(16x^3)$
4. $\cos(\frac{2}{x})$

Solution. 1. Differentiating $\ln(5x)$

Let

$$y = \ln(5x), \quad u = 5x$$

so that $y = \ln u$.

We compute

$$\begin{aligned} \frac{dy}{du} &= \frac{1}{u} && \text{(using the derivative of } \ln u) \\ \frac{du}{dx} &= 5 && \text{(using the derivative of } 5x) \end{aligned}$$

Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \frac{1}{u} \cdot 5 \\ &= \frac{5}{5x} \\ &= \frac{1}{x} \end{aligned}$$

2. Differentiating e^{-20x^2}

Let

$$y = e^{-20x^2}, \quad u = -20x^2$$

so that $y = e^u$.

We compute

$$\begin{aligned} \frac{dy}{du} &= e^u && \text{(using the derivative of } e^u) \\ \frac{du}{dx} &= -40x && \text{(using the derivative of } -20x^2) \end{aligned}$$

Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= e^u(-40x) \\ &= -40xe^{-20x^2}\end{aligned}$$

3. Differentiating $\sin(16x^3)$

Let

$$y = \sin(16x^3), \quad u = 16x^3$$

so that $y = \sin u$.

We compute

$$\begin{aligned}\frac{dy}{du} &= \cos u && \text{(using the derivative of } \sin u) \\ \frac{du}{dx} &= 48x^2 && \text{(using the derivative of } 16x^3)\end{aligned}$$

Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \cos u \cdot 48x^2 \\ &= 48x^2 \cos(16x^3)\end{aligned}$$

4. Differentiating $\cos\left(\frac{2}{x}\right)$

Let

$$y = \cos\left(\frac{2}{x}\right), \quad u = \frac{2}{x}$$

so that $y = \cos u$.

We compute

$$\begin{aligned}\frac{dy}{du} &= -\sin u && \text{(using the derivative of } \cos u) \\ \frac{du}{dx} &= -\frac{2}{x^2} && \text{(using the derivative of } 2x^{-1})\end{aligned}$$

Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= (-\sin u) \left(-\frac{2}{x^2}\right) \\ &= \frac{2 \sin\left(\frac{2}{x}\right)}{x^2}\end{aligned}$$

□

Exercise 2. Evaluate the following indefinite integrals.

1. $\int \sin^2 x \, dx$

2. $\int \cos^2 x \, dx$

Solution. 1. Integrating $\sin^2 x$

We use the identity

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

Therefore

$$\begin{aligned}\int \sin^2 x \, dx &= \int \frac{1 - \cos(2x)}{2} \, dx \\&= \frac{1}{2} \int 1 \, dx - \frac{1}{2} \int \cos(2x) \, dx \\&= \frac{1}{2}x - \frac{1}{2} \cdot \frac{1}{2} \sin(2x) + c \\&= \frac{x}{2} - \frac{1}{4} \sin(2x) + c\end{aligned}$$

2. Integrating $\cos^2 x$

We use the identity

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

Therefore

$$\begin{aligned}\int \cos^2 x \, dx &= \int \frac{1 + \cos(2x)}{2} \, dx \\&= \frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos(2x) \, dx \\&= \frac{1}{2}x + \frac{1}{2} \cdot \frac{1}{2} \sin(2x) + c \\&= \frac{x}{2} + \frac{1}{4} \sin(2x) + c\end{aligned}$$

□

Exercise 3. For each integral below, state a suitable substitution before evaluating the indefinite integral.

1. $\int \frac{5}{5x+2} \, dx$

2. $\int e^{-20x} \, dx$

3. $\int \sin(16x) \, dx$

4. $\int \cos(2x) \, dx$

Solution. 1. Integrating $\int \frac{5}{5x+2} \, dx$

We use the substitution

$$u = 5x + 2$$

so that $du = 5 \, dx$.

Then

$$\begin{aligned}\int \frac{5}{5x+2} \, dx &= \int \frac{1}{u} \, du \\&= \ln |u| + c \\&= \ln |5x+2| + c\end{aligned}$$

2. Integrating $\int e^{-20x} dx$

We use the substitution

$$u = -20x$$

so that $du = -20 dx$ and $dx = -\frac{1}{20} du$.

Then

$$\begin{aligned}\int e^{-20x} dx &= -\frac{1}{20} \int e^u du \\ &= -\frac{1}{20} e^u + c \\ &= -\frac{1}{20} e^{-20x} + c\end{aligned}$$

3. Integrating $\int \sin(16x) dx$

We use the substitution

$$u = 16x$$

so that $du = 16 dx$ and $dx = \frac{1}{16} du$.

Then

$$\begin{aligned}\int \sin(16x) dx &= \frac{1}{16} \int \sin u du \\ &= -\frac{1}{16} \cos u + c \\ &= -\frac{1}{16} \cos(16x) + c\end{aligned}$$

4. Integrating $\int \cos(2x) dx$

We use the substitution

$$u = 2x$$

so that $du = 2 dx$ and $dx = \frac{1}{2} du$.

Then

$$\begin{aligned}\int \cos(2x) dx &= \frac{1}{2} \int \cos u du \\ &= \frac{1}{2} \sin u + c \\ &= \frac{1}{2} \sin(2x) + c\end{aligned}$$

□